# PLASMA PHYSICS & TOKAMAK CONFINEMENT — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

*Diagram-based questions on Plasma Physics & Tokamak Confinement*

## Question 1

10.0 points

Examine the toroidal and poloidal magnetic field geometry in the tokamak cross-section shown in the diagram below. Discuss how these fields collaborate to confine plasma, and identify any potential areas of instability within the configuration.

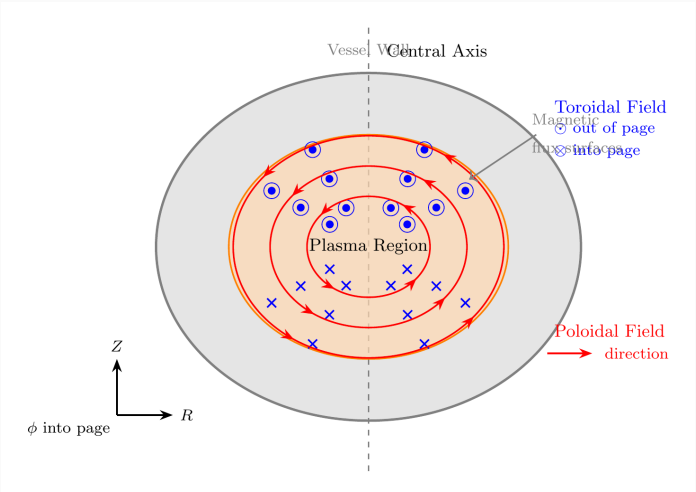


Fig. 1. A tokamak cross-section showing the toroidal and poloidal magnetic field geometry. The diagram should depict the toroidal magnetic field lines wrapping around the torus in the horizontal direction, and the poloidal magnetic field lines looping in the vertical direction around the plasma. The central plasma column should be shown, with the magnetic field lines clearly labeled as 'Toroidal Field' and 'Poloidal Field'. Indicate the direction of the fields with arrows. Highlight potential areas of instability, such as regions where the field lines might intersect or where the configuration might be weak.

## Question 2

10.0 points

Interpret the Lawson criterion diagram below that displays the triple product $nT\tau$ versus temperature. Explain how this diagram is used to evaluate the feasibility of fusion reactions.

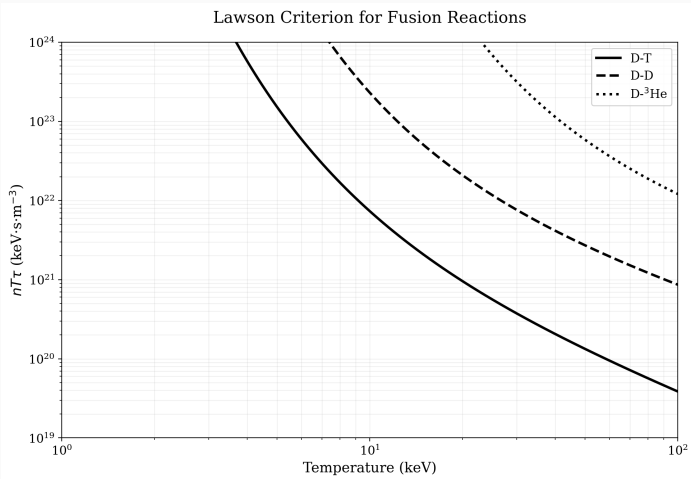


Fig. 2. Lawson criterion plot showing the triple product nTτ versus temperature for evaluating fusion reaction feasibility. The plot includes regions indicating where different fusion reactions (such as D-T, D-D, and D-He3) are feasible, with labeled axes for temperature and nTτ.

## Question 3

10.0 points

Examine the diagram below illustrating magnetic field curvature and grad-B drift for a charged particle in a tokamak. Determine how these drifts impact particle confinement and suggest mitigation strategies.

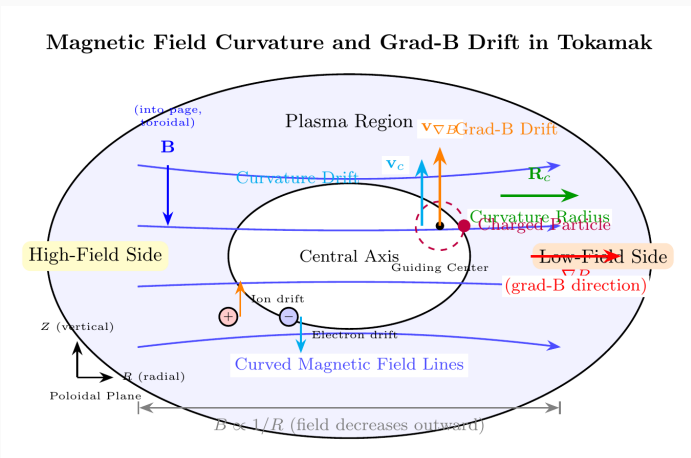


Fig. 3. WR

## Question 4

10.0 points

Examine the $q$-profile (safety factor) and magnetic flux surfaces in the tokamak shown in the diagram below. Discuss how the shape of the $q$-profile influences plasma stability and confinement.

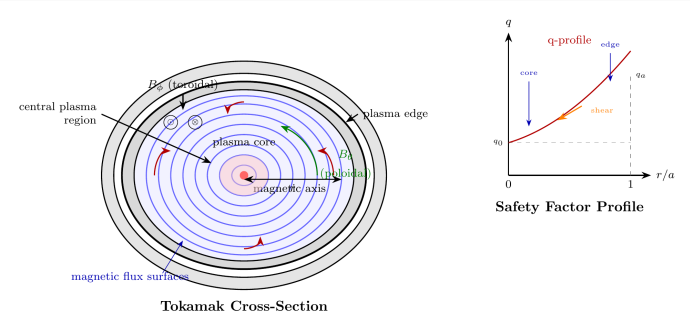


Fig. 4. A cross-section of a tokamak showing magnetic flux surfaces and the q-profile (safety factor). The diagram should include concentric magnetic flux surfaces labeled with increasing radii from the center. The q-profile should be represented as a curve or graph overlaying the flux surfaces, illustrating how the safety factor changes with radius. Label key components such as the plasma core, magnetic axis, and edge of the plasma. The diagram should help visualize the influence of the q-profile on plasma stability and confinement.

## Question 5

10.0 points

Examine the diagram below that illustrates the plasma instability growth rate as a function of wave number. Discuss the implications of this relationship for tokamak operation and stability control.

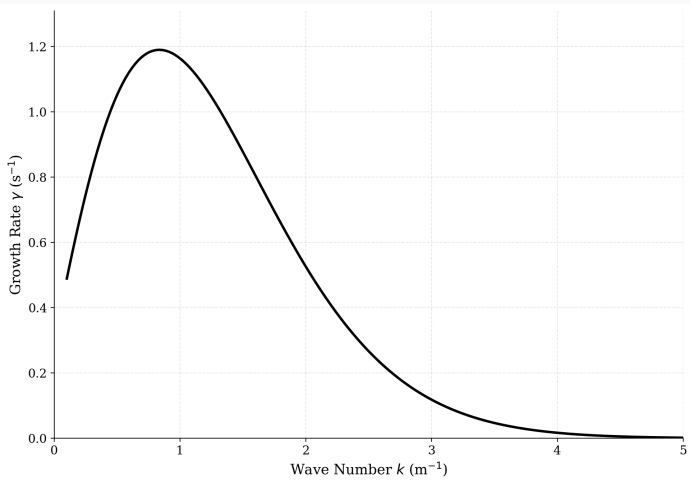


Fig. 5. A plot illustrating plasma instability growth rate as a function of wave number. The x-axis represents the wave number, and the y-axis represents the growth rate. The plot should show a typical curve where the growth rate increases with wave number up to a certain point, then decreases, indicating the most unstable wave number. This visual helps explain the implications for tokamak operation and stability control.